

Arjuna JEE 2027

Maths by Amarnath Anand Sir

Quadratic Equations

DPP: 3

Total Duration - 35 Min

Q1 The value of $3 + \frac{1}{4 + \frac{1}{3 + \frac{1}{4 + \frac{1}{3 + \dots \infty}}}}$ is equal to

- (A) $1.5 + \sqrt{3}$
 (B) $2 + \sqrt{3}$
 (C) $3 + 2\sqrt{3}$
 (D) $4 + \sqrt{3}$

Q2 The values of k for which $2x^2 + kx + 2 = 0$ has real roots, are:

- (A) $\{k \in \mathbb{R} : -4 \leq k \leq 4\}$
 (B) $\{k \in \mathbb{R} : k \geq 4\}$
 (C) $\{k \in \mathbb{R} : k \geq 4 \text{ or } k \leq -4\}$
 (D) None of these

Q3 If the equation $ax^2 - 5x + c = 0$ has 10 as the sum of the roots and also as the product of the roots, then

- (A) $a = \frac{1}{2}, c = 5$
 (B) $a = 2, c = 3$
 (C) $a = 5, c = \frac{1}{2}$
 (D) $a = 3, c = 2$

Q4 If a, b, c are positive real numbers, then the number of real roots of the equation

$$ax^2 + b|x| + c = 0 \text{ is}$$

- (A) 0
 (B) 2
 (C) 4
 (D) 1

Q5 If α and β are the roots of the equation $x^2 - px + 16 = 0$, such that $\alpha^2 + \beta^2 = 9$, then the value of p is

- (A) $\pm\sqrt{6}$
 (B) $\pm\sqrt{41}$
 (C) ± 8
 (D) ± 7

Q6 Let α and β are the roots of the equation $x^2 + x + 1 = 0$ then

- (A) $\alpha^2 + \beta^2 = 4$

$$(B) (\alpha - \beta)^2 = 3$$

$$(C) \alpha^3 + \beta^3 = 2$$

$$(D) \alpha^4 + \beta^4 = 1$$

Q7 If α and β are the roots of the equation $x^2 + 6x + \lambda = 0$ and $3\alpha + 2\beta = -20$, then $\lambda =$

- (A) -8
 (B) -16
 (C) 16
 (D) 8

Q8 A value of k for which the quadratic equation $x^2 - 2x(1 + 3k) + 7(2k + 3) = 0$ has equal roots is

- (A) 1
 (B) 2
 (C) 3
 (D) 4

Q9 If the product of the roots of the equation $(a + 1)x^2 + (2a + 3)x + (3a + 4) = 0$ be 2, then the sum of roots is

- (A) 1
 (B) -1
 (C) 2
 (D) -2

Q10 If $\sin \theta, \cos \theta$ are the roots of $ax^2 + bx + c = 0$, then:

- (A) $a^2 = b^2 + 2ab$
 (B) $b^2 = a^2 + 2ab$
 (C) $a^2 = b^2 + 2ac$
 (D) $b^2 = a^2 + 2ac$

Q11 If α and β are roots of quadratic equation $x^2 - 4x + 2 = 0$, then value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} =$

Q12 The value of 'a' for which one root of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other is

- (A) $-\frac{1}{3}$
 (B) $\frac{2}{3}$
 (C) $-\frac{2}{3}$
 (D) None of these



- Q13** If α and β be the roots of the equation $(x - a)(x - b) = c$ and $c \neq 0$, then roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are
(A) a and c
(B) b and c
(C) a and b
(D) $a + b$ and $b + c$

- Q14** If $(1 - p)$ is a root of quadratic equation $x^2 + px + (1 - p) = 0$ then its root are-
(A) $0, -1$ (B) $-1, 1$
(C) $0, 1$ (D) $-1, 2$

- Q15** If α, β are roots of the equation $4x^2 + 3x + 7 = 0$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ is equal to
(A) $7/3$
(B) $-7/3$
(C) $3/7$
(D) $-3/7$

- Q16** If α, β are the roots of a quadratic equation $x^2 - 3x + 5 = 0$, then the equation where roots are $(\alpha^2 - 3\alpha + 7)$ and $(\beta^2 - 3\beta + 7)$ is
(A) $x^2 + 4x + 1 = 0$
(B) $x^2 - 4x + 4 = 0$
(C) $x^2 - 4x - 1 = 0$
(D) $x^2 + 3x + 3 = 0$



Answer Key

Q1 (A)
Q2 (C)
Q3 (A)
Q4 (A)
Q5 (B)
Q6 (C)
Q7 (B)
Q8 (B)

Q9 (B)
Q10 (D)
Q11 6
Q12 (B)
Q13 (C)
Q14 (A)
Q15 (D)
Q16 (B)



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